

AI-Powered Virtual Assistant for the NUS MSBA Website

Sneha More

National University of Singapore

Abstract—Chatbots have emerged as a key application of artificial intelligence in higher education, providing scalable, on-demand access to information and enhancing student engagement. This project presents the design and implementation of a domain-specific chatbot for the NUS Master of Science in Business Analytics (MSBA) programme, built on a Retrieval-Augmented Generation (RAG) architecture using Google Vertex AI Agent Builder. The system integrates official programme documents, including brochures and FAQs, to deliver accurate, citation-supported responses to prospective students’ queries. Development encompassed structured content preprocessing, embedding generation, retrieval configuration, and fallback response design. Evaluation, based on a curated set of 50 representative queries spanning admissions, curriculum, tuition, and programme details, assessed accuracy, clarity, traceability, and fallback appropriateness. Iterative refinement improved accuracy from 76% to 88%, particularly for fact-based questions. While results indicate strong performance in real-world testing, limitations remain in fuzzy matching and the lack of built-in real-time analytics. The work demonstrates the feasibility and effectiveness of deploying RAG-based chatbots in higher education, offering a roadmap for enhanced retrieval robustness, user experience, and monitoring capabilities.

Index Terms—Retrieval-Augmented Generation (RAG), Chatbot, Higher Education, Google Vertex AI, Knowledge Retrieval, Natural Language Processing, Conversational AI.

I. INTRODUCTION

In today’s digital-first environment, academic institutions are increasingly adopting advanced technologies to enhance engagement, streamline administrative workflows, and provide timely, accurate information to prospective students. For globally recognised programmes such as the Master of Science in Business Analytics (MSBA) at the National University of Singapore (NUS), the rising interest from both local and international applicants has resulted in a significant increase in queries related to application procedures, programme structure, tuition fees, and post-graduation opportunities.

Traditionally, these inquiries have been addressed through static FAQs on the official website or via direct email correspondence. While effective to an extent, these methods require considerable staff resources and often lead to delays. Ensuring the accuracy and consistency of responses presents an additional challenge, particularly when the information sought is dynamic or context-specific.

Recent advancements in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) present an opportunity to overcome these limitations. By combining the natural language capabilities of LLMs with curated, institution-approved content, it is possible to develop conversational AI systems that provide accurate, context-aware, and human-

like responses, while maintaining strict adherence to verified information.

This project explores the development of an AI-powered conversational assistant designed specifically for the NUS MSBA programme, built using Google Cloud’s Vertex AI Agent Builder. The proof-of-concept integrates official programme materials to deliver instant, traceable, and institutionally accurate responses to prospective student queries. While testing has demonstrated promising results, full deployment will require clearance through NUS IT’s Cloud Assessment Process and, if applicable, review by the Solution Review Board (SRB). NUS IT has also indicated that future integration with the university’s in-house AI-Know chatbot may be necessary for production use.

A. Problem Context and Challenges

The NUS MSBA programme receives a high volume of queries from prospective students each year. Although the official website and brochures contain comprehensive information, navigating through extensive documents to find specific answers can be challenging for applicants. As a result, repetitive questions such as “What are the admission requirements?”, “How much is the tuition fee?”, or “When is the application deadline?” are frequently directed to administrative staff, increasing workload and creating the potential for delays or inconsistencies in communication.

Existing solutions have notable limitations. Traditional rule-based chatbots or static web forms are unable to accommodate varied phrasings or nuanced questions, while generic AI assistants risk producing inaccurate or outdated information if not grounded in trusted sources. The central challenge lies in creating a system that combines the fluency of LLMs with the factual reliability of institutionally verified content.

This project proposes a **proof-of-concept RAG-based chatbot** within the Google Vertex AI ecosystem to address these challenges by:

- Delivering accurate, scalable, and secure responses.
- Reducing administrative workload.
- Ensuring all outputs are aligned with official MSBA content.

Given these limitations, it becomes critical to understand why a more advanced approach is necessary, and how it aligns with the broader needs of academic institutions such as NUS.

B. Rationale and Research Motivation

The decision-making process for prospective graduate students involves extensive information gathering, comparison across institutions, and clarity around expectations. Academic programs like the NUS Master of Science in Business Analytics (MSBA),

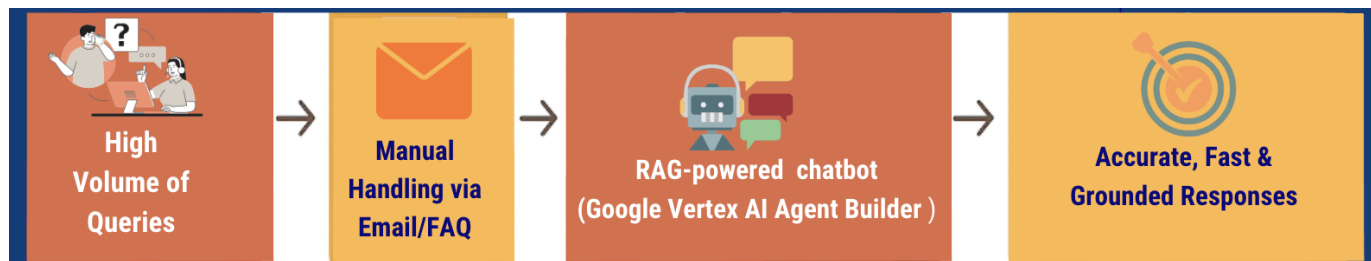


Figure 1. Problem to Solution Overview

which operate in a highly competitive and international context, must ensure that accurate and relevant information is readily accessible to prospective applicants. However, despite having well-structured web pages and downloadable brochures, students often face challenges in locating specific answers to their questions. This friction can lead to frustration, discourage highly qualified candidates, or result in repeated inquiries that increase the administrative workload of admissions teams.

Advances in generative artificial intelligence, particularly in **Large Language Models (LLMs)**, have enabled the development of conversational agents capable of responding to natural language queries with fluency and contextual understanding. While these capabilities make LLMs attractive for student engagement, they also introduce the well-documented risk of “hallucination,” where plausible but incorrect or unverifiable answers are produced. This limitation underscores the importance of grounding AI-generated responses in reliable data sources.

The Retrieval-Augmented Generation (RAG) framework directly addresses this challenge by combining the linguistic flexibility of LLMs with the factual grounding of a curated document corpus. In this architecture, the chatbot retrieves the most relevant pieces of information from its knowledge base and uses only that retrieved content to generate responses. This ensures that answers remain both conversational and traceable to verifiable sources, providing a level of transparency and reliability essential in an academic context.

In the context of the MSBA chatbot, the RAG process begins when a user submits a query, which is matched against the indexed knowledge base containing program brochures, FAQs, and other official materials. Relevant content is then retrieved and augmented with the query before being passed to the LLM for response generation. The system delivers the final answer to the user along with citations that reference the original source documents. This workflow, shown in Figure 2, reflects a balance between conversational engagement and institutional trust.

The workflow is as below:

- **User Query Submission** – The user enters a question into the chatbot interface.
- **Content Retrieval** – The system searches a curated, indexed knowledge base containing program brochures, FAQs, and other official materials.
- **Augmentation & Generation** – Retrieved content is paired with the query and passed to the LLM for response generation.
- **Cited Response Delivery** – The chatbot responds with a

conversational answer and inline citations linking to the original sources.

The choice of **Google Vertex AI Agent Builder** for this project was guided by its secure, enterprise-grade environment, integrated support for document grounding, and no-code interface that allows non-technical staff to maintain and update the chatbot after deployment. Its seamless integration with Google Cloud Storage supports versioned, scalable updates to the knowledge base, while domain-restricted embedding ensures that the chatbot remains secure and cannot be deployed on unauthorized websites.

Criteria	Google Vertex AI Agent Builder	Open-Source Framework
Security & Compliance	Enterprise-grade, ISO/GDPR compliant	Depends on implementation & hosting
Ease of Maintenance	No-code UI for content updates	Requires technical expertise
Integration	Native Google Cloud Storage support	Flexible, but manual integration
Scalability	Auto-scalable infrastructure	Depends on hosting setup
Customisation	High domain-specific flexibility	Full control, but high development effort

Table 1. Comparison of Google Vertex AI Agent Builder and Open-Source Conversational AI Frameworks

Overall, this project serves as both a proof of concept and a practical implementation of how institutions like NUS can responsibly adopt generative AI technologies to modernize their outreach and support services while upholding institutional trust and content integrity.

This proof-of-concept also serves as an opportunity to evaluate the trade-offs between open-source experimentation and managed enterprise platforms. While open-source frameworks offer flexibility and control, they demand significant technical expertise and maintenance resources. In contrast, managed platforms like Google Vertex AI Agent Builder prioritize governance, compliance, and operational simplicity, making them more suitable for long-term institutional adoption.

Beyond its technical contribution, the project underscores the importance of responsible AI integration in higher education. By grounding responses in verified programme materials and embedding mechanisms for transparency, the system not only improves efficiency but also builds trust with prospective students. In doing so, it highlights how universities can leverage

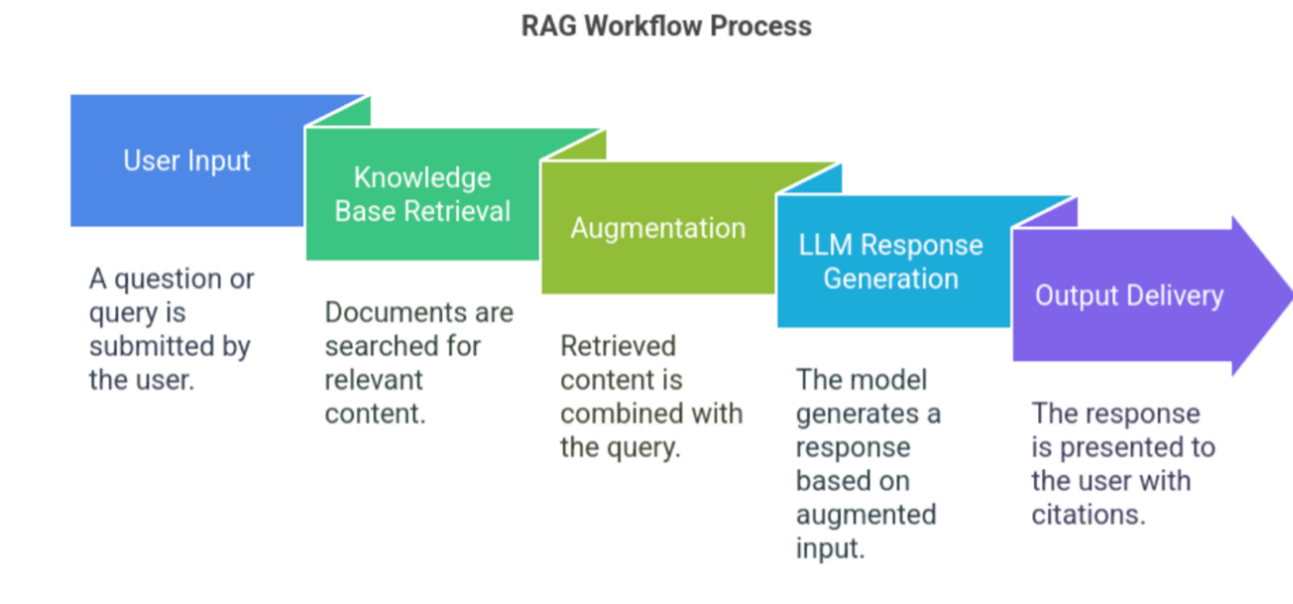


Figure 2. Retrieval-Augmented Generation (RAG) Workflow Process [1]

cutting-edge AI while safeguarding accuracy, accountability, and institutional values.

II. LITERATURE REVIEW AND RELATED WORK

A. Evolution of Conversational AI

In recent years, conversational AI has evolved from experimental tools into essential enablers of digital transformation in higher education. Universities are increasingly turning to chatbots to improve student engagement, streamline administrative processes, and deliver timely, accurate information to stakeholders.

Early academic chatbots predominantly relied on rule-based logic, keyword matching, and pre-scripted dialogue flows. These systems were functional for answering frequently asked questions but lacked scalability, adaptability, and contextual understanding. They struggled with varied phrasing, multi-intent queries, and dynamic context handling leading to rigid, repetitive responses. For tech-savvy prospective students expecting instant, personalized, and natural interactions, this was inadequate. While predictable and secure, these systems often failed to capture the variability of natural language and could not adapt to evolving programme content. As these systems scaled, they became brittle requiring frequent manual updates to handle new query formats, accelerating the shift towards AI-powered conversational agents.

The arrival of transformer-based Large Language Models (LLMs) such as GPT-3, GPT-4, and PaLM 2 transformed the capabilities of conversational systems, enabling them to understand and respond to diverse, context-rich queries. However, their purely generative nature introduces risks, most notably the production of factually incorrect or “hallucinated” responses. In high-stakes academic environments, where accuracy and credibility are non-negotiable, this risk is unacceptable.

To address this, Retrieval-Augmented Generation (RAG) has gained prominence. RAG combines the generative capabilities of LLMs with document retrieval from curated knowledge bases, ensuring responses are both fluent and factually grounded. This hybrid method reduces hallucination risk, improves factual accuracy, and enables traceability via citations to original sources. Academic studies consistently show that RAG-based chatbots outperform both static FAQ pages and purely generative systems in accuracy, trustworthiness, and user satisfaction.

Approach	Strengths	Limitations
Rule-based	Predictable, secure, low computational cost	Rigid, high maintenance, poor scalability
Pure LLM	Flexible, natural, adaptable to varied queries	Prone to hallucination, less controllable
RAG-based	Accurate, grounded, balances fluency and reliability	Requires regular document updates, retrieval infrastructure

Table 2. Comparative Overview of Conversational AI Approaches

Having established the evolution from rule-based to RAG-based approaches, it is important to examine the specific tools and frameworks that enable these systems in practice.

B. Tools and Frameworks for RAG

Several open-source RAG frameworks including LangChain and LlamaIndex have become popular for building custom knowledge-grounded assistants, offering flexibility and full control over deployment. However, they require substantial development expertise, infrastructure, and ongoing maintenance. In contrast, managed cloud platforms such as Google Vertex AI Search and Conversation or Microsoft Azure AI Bot Service provide low-code environments with built-in scalability, security, and compliance, though their adoption within

universities often depends on IT governance approval and integration with legacy systems. **Open-Source Exploration:** During prototyping, the project evaluated open-source RAG stacks including **LangChain, LlamaIndex, and Pinecone**:

- LangChain offered modular chaining of retrieval, memory, and agent logic, enabling flexible pipeline design.
- LlamaIndex excelled in parsing diverse formats, intelligent text chunking, and building custom retrieval indices.
- Pinecone provided fast, scalable vector search with strong performance for similarity queries.

These tools enabled rapid experimentation and fine-grained control over retrieval pipelines. However, challenges emerged:

- High technical barrier – required advanced programming skills, making them inaccessible for non-technical staff.
- Operational overhead – managing hosting, billing, and scaling (especially for Pinecone) added complexity.
- Data security concerns – compliance and governance required additional safeguards.
- Maintenance burden – embedding and sustaining these stacks within institutional websites proved cumbersome.
- While effective for proof-of-concept development, they were not optimal for long-term production deployment.

Transition to Google Vertex AI Agent Builder: To overcome these limitations, the project adopted Google Vertex AI Agent Builder, a managed, enterprise-grade RAG platform. Key advantages include:

- Integrated workflow for ingestion, chunking, embedding, indexing, and querying within the Google Cloud ecosystem.
- Secure storage via Google Cloud Storage (GCS) with Identity and Access Management (IAM) controls.
- Low-code UI enabling faculty and administrative staff to update the knowledge base and refine responses without coding.
- Built-in grounding and citation features to link responses to verified sources.
- Seamless web integration with domain whitelisting for security.
- Enterprise-grade compliance and scalability as part of Google Cloud.

While customization options are more limited than with open-source frameworks, the trade-off is justified by its operational simplicity, governance features, and maintainability for a university setting.

Table 3 summarizes the **evaluation of the key tools and platforms considered during this project**.

Tool/Platform	Advantages	Limitations	Suitability for MSBA Chatbot
LlamaIndex	Easy document parsing and indexing	Requires coding, no built-in UI	Good for prototyping
LangChain	Flexible modular workflows	Steep learning curve, not no-code	Experimental use only
Pinecone	Fast, scalable vector search	Separate billing and hosting	Not ideal for internal ops
Vertex AI Agent Builder	End-to-end managed platform, secure, UI-driven	Limited customization but suitable trade-off	Ideal for production use

Table 3. Evaluation of RAG-Based Chatbot Tools

Having reviewed both open-source and managed platforms, it is equally important to consider how these approaches are being applied within higher education, and specifically within NUS.

C. Application in Higher Education Context

Within the National University of Singapore (NUS), the AI-Know chatbot represents a centralized institutional effort to provide AI-powered assistance across faculties. Leveraging an in-house platform offers advantages in compliance, data security, and integration. However, domain-specific, custom solutions like the one developed in this project enable faster prototyping and more targeted optimization during early proof-of-concept phases, before potential integration into enterprise-wide systems. The present project sits at the intersection of these approaches: it applies a managed cloud RAG solution (Google Vertex AI Agent Builder) to a highly domain-specific context, the MSBA programme while maintaining alignment with institutional initiatives such as AI-Know. This hybrid approach reflects broader trends in higher education AI adoption, where controlled experimentation with external platforms informs the design and requirements for future in-house deployments.

III. METHODOLOGY

A. Overview

This project followed a structured methodology to design, configure, and evaluate an AI-powered conversational assistant for the NUS MSBA programme. The process combined data preparation, corpus creation, system configuration within Google Cloud’s Vertex AI Agent Builder, and integration with the MSBA website. While the technical work demonstrated a functional proof-of-concept, the methodology was designed with maintainability, security, and scalability in mind, ensuring that the solution could be adapted for future institutional integration.

The methodology aligns closely with the Generative AI Workflow on Vertex AI illustrated in Figure 3. This workflow demonstrates the sequence from user prompt input within the Vertex AI Studio UI, through the application of responsible AI and safety mechanisms, to processing via foundation models

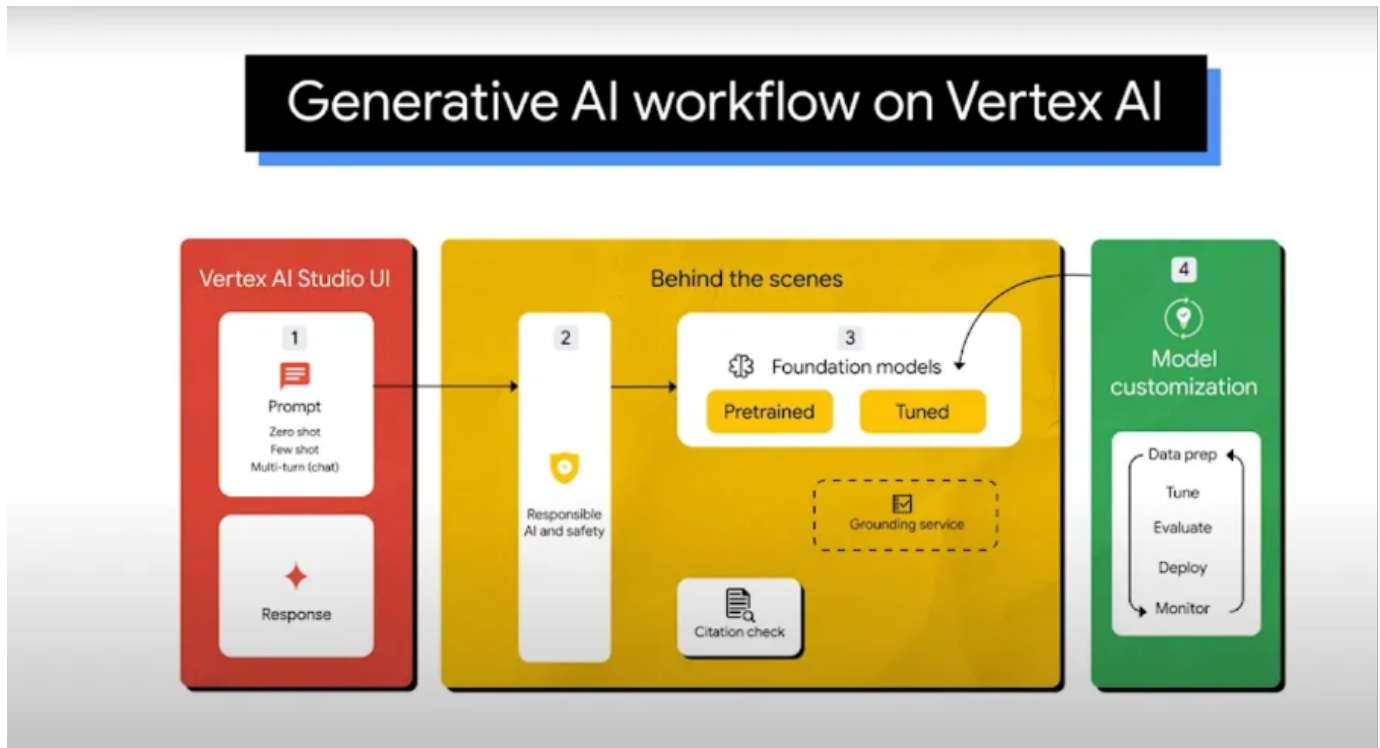


Figure 3. GenAI workflow on Vertex AI [2]

(pretrained and tuned), and finally into model customization stages such as data preparation, tuning, evaluation, deployment, and monitoring. This structured flow ensures that every response generated is both contextually relevant and compliant with institutional standards, while allowing for ongoing optimization during future iterations.

B. Document Preparation and Ingestion

Following the workflow outlined in Figure 2, the first step in the implementation involved preparing the knowledge base for ingestion into Vertex AI Agent Builder. The chatbot's corpus was built entirely from institution-approved MSBA-specific materials to ensure both accuracy and compliance. Three primary sources formed the foundation of the corpus: the **MSBA brochure**, which contained detailed curriculum information, fee structures, and module descriptions (**Appendix A, Figure A1**); the **MSBA FAQ document**, which addressed recurring prospective student questions and selected text extracted from the official **MSBA website**, used to fill informational gaps and ensure currency.

Prior to ingestion, each document underwent **preprocessing** to remove formatting inconsistencies, superfluous symbols, irregular line breaks, and unnecessary metadata. This step ensured uniformity in the subsequent **chunking and embedding** phases. The cleaned documents were uploaded to a secure **Google Cloud Storage (GCS) bucket** linked to the chatbot project, ensuring both controlled access and smooth integration with Vertex AI Agent.

C. Corpus Creation and Embedding Workflow

Once uploaded, the documents were processed through Vertex AI Agent Builder's default **Layout Parser** to identify document structures. For PDF files, the parser was overridden with an **OCR-specific** configuration to correctly interpret scanned documents and complex layouts (**Appendix A, Figure A2**).

Advanced chunking was applied, segmenting content into 500-token chunks with ancestor headings retained to preserve contextual hierarchy. This granular segmentation enabled more precise retrieval and semantic matching between user queries and document segments.

In addition, table and image annotations were enabled, allowing the model to understand and retrieve information from visual and tabular elements. The processed chunks were embedded using Google's proprietary language model and stored within a managed vector database which served as the semantic search backbone for the Retrieval-Augmented Generation (RAG) pipeline. This approach reduced the risk of hallucinations by constraining the model's responses to relevant, pre-approved content.

D. Application Configuration in Vertex AI Agent Builder

The chatbot's operational behaviour was defined through several key configurations within Vertex AI Agent Builder. The model context was set to restrict responses strictly to MSBA-related topics, preventing the generation of unrelated or speculative content. A professional yet approachable tone was applied through the personality settings, with verbosity maintained at

a medium level to balance completeness with conciseness. (Appendix A, Figure A3)

Fallback mechanisms (Appendix A, Figure A4) were implemented to manage cases where no relevant corpus content was found. Custom fallback messages politely informed users of the limitation, while selective Large Language Model fallback was enabled to address generic queries such as contact details without compromising accuracy. In all cases, safeguards such as prompt grounding and model context restrictions were applied to mitigate hallucination risks.

Citation transparency (Appendix A, Figure A4) was also prioritised. When a response was generated from corpus content, the chatbot displayed both the source document and the specific text snippet used. Furthermore, **conversational logging** was enabled to capture all queries, responses, and associated metadata, enabling post-session analysis of chatbot performance (Appendix A, Figure A5). Complementing this, a thumbs up/down **feedback mechanism** allowed users to rate responses in real time, with ratings stored alongside logs for performance tracking and continuous improvement.

E. Domain and Widget Integration

To enable access through the MSBA website, the project will need to complete domain verification via Google Search Console, confirming ownership of the msba.nus.edu.sg domain. A customised Google Messenger widget is configured to align with the programme’s branding, including the application of the MSBA colour scheme, a welcoming greeting message, and conversation reset rules. The widget script provided by Agent Builder will be shared with the university’s web development team for placement on the site.

The verified domain needs to be whitelisted within Vertex AI Agent Builder to prevent unauthorised deployments. While not fully implemented in this proof-of-concept stage, the project also explored the possibility of enabling dynamic content crawling through Vertex AI’s advanced website indexing feature, a potential enhancement for future deployment following institutional review and approval. The complete end-to-end configuration, from user interaction to secure deployment, is illustrated in the **MSBA Chatbot Technical Architecture** diagram, which maps the system’s layers and components described above.

IV. SYSTEM ARCHITECTURE

The MSBA Chatbot is implemented using a layered architecture that ensures clear separation of responsibilities, scalability, and secure integration with institutional systems. The architecture, shown in **Figure 4**, is composed of five layers: User Interaction, Processing, Retrieval, Response Generation, and Platform Services.

The User Interaction Layer consists of the MSBA website, where the chatbot is embedded as a Google Vertex AI Messenger widget. This allows prospective students to interact with the system in a familiar web interface without requiring additional installations.

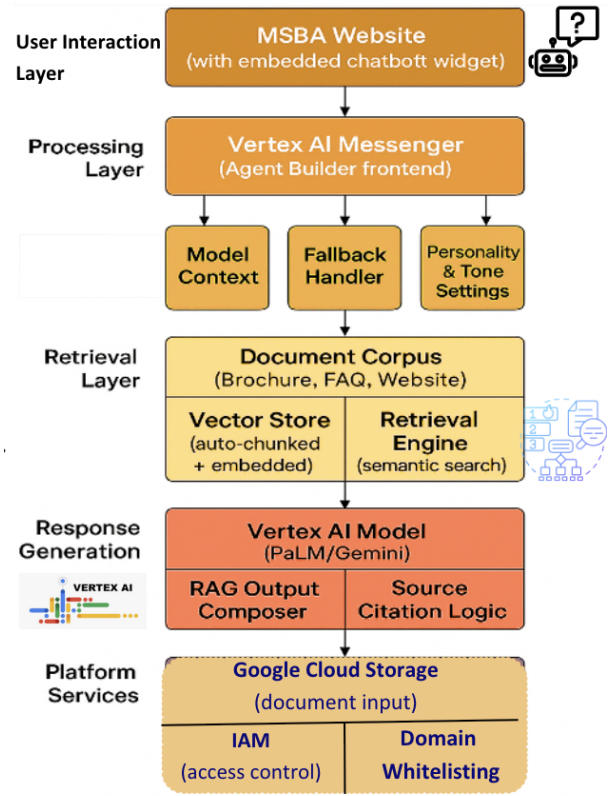


Figure 4. System Architecture Diagram

The Processing Layer operates in two tiers. First, the Vertex AI Messenger interface, powered by Agent Builder, receives user input and routes it into the system. Second, the chatbot applies Model Context rules to restrict responses to MSBA-related content, a Fallback Handler for managing queries outside the corpus, and Personality & Tone Settings to maintain a professional yet approachable communication style.

The Retrieval Layer forms the semantic search backbone. It consists of the Document Corpus comprising the MSBA brochure, FAQs, and curated website content stored in a structured format. The Vector Store holds embeddings of these documents, generated through Vertex AI’s auto-chunking process, while the Retrieval Engine uses semantic matching to identify the most relevant document segments for a given query.

The Response Generation Layer leverages Google’s Vertex AI large language models, including PaLM and Gemini. Once relevant context is retrieved, the RAG Output Composer constructs the final response, and the Source Citation Logic appends transparent attributions to each answer, enabling users to verify the origin of the information.

The Platform Services Layer provides the supporting infrastructure. Google Cloud Storage (GCS) serves as the secure repository for document ingestion, while IAM controls access permissions. Domain Whitelisting ensures that only authorised web domains can deploy the chatbot widget.

This structured architecture not only enables a functional proof-of-concept but also positions the system for seamless inte-

gration with NUS IT’s AI-Know framework, future multilingual support, and advanced analytics capabilities.

V. TESTING AND EVALUATION

A. Testing Methodology

A systematic testing process was conducted using a set of approximately fifty representative queries compiled with the MSBA Business Analytics Centre (BAC). The queries reflected real prospective-student information needs across admissions, curriculum, tuition fees, application processes, career services, and general programme details. The distribution of these queries is shown in Figure 5. For each item, the record included the original input, the chatbot’s output, any source citation, observed issues, reviewer feedback, and resolution status. Evaluation considered accuracy (alignment with official programme content), clarity (readability and professional tone), traceability (presence of valid citations or links), and fallback behavior (usefulness when relevant content was not found). Robustness was explored with alternate phrasings and common misspellings (e.g., “tution” for “tuition”). Conversation logging and the thumbs-up/thumbs-down feature provided additional signals for review.

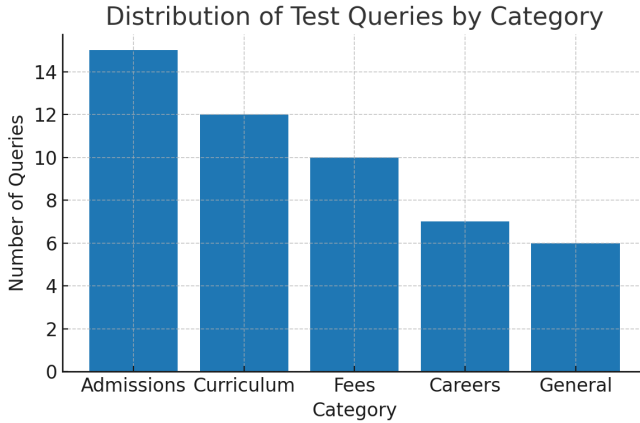


Figure 5. Distribution of test queries by category.

B. Key Findings

Across the test set, the assistant performed strongly on fact-based questions grounded in MSBA materials. Direct prompts such as “What are the admission requirements?” consistently returned accurate answers linked to official pages. Formatting artefacts from brochure PDFs occasionally produced spacing issues in currency values (for example, “S 64,800”); prioritizing cleaned FAQ and website content mitigated this. Review of conversation logs indicated that highly rated responses tended to synthesize information from multiple sources, while lower ratings commonly reflected ambiguous queries or residual formatting problems. The fallback configuration proved effective for subjective questions (e.g., “Is the MSBA programme worth it?”), directing users to appropriate channels rather than speculating. A summary of observed outcomes from internal testing is shown in Figure 6.

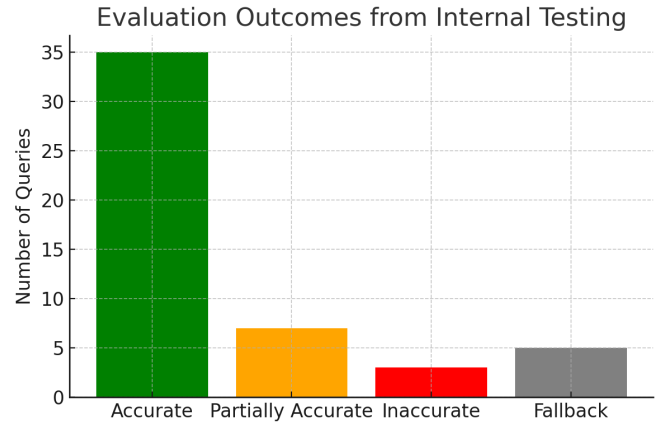


Figure 6. Evaluation outcomes from internal testing.

C. Proposed Metrics for Formal Evaluation

While the approach above provided immediate feedback during development, it was not a controlled or large-scale evaluation. The results should be regarded as indicative rather than statistically significant. For any pilot or production phase, a formal evaluation framework is recommended, with clearly defined measures for factual reliability, retrieval quality, responsiveness, and user satisfaction. Table 4 outlines proposed metrics and definitions that align with established evaluation practice for information-seeking dialogue systems.

Metric	Description
Accuracy	Proportion of responses that are factually correct against official sources
Grounding Fidelity	Percentage of responses with explicit citation or source linkage
Precision@k	Proportion of the top-k retrieved chunks that are relevant to the query
Fallback Appropriateness	Share of no-match responses that provide helpful next steps (e.g., contact or link)
Latency (p95/p99)	Response time at the 95th/99th percentiles during interactive use
User Satisfaction	Positive rating rate (thumbs-up or short post-interaction survey)

Table 4. Definitions of planned metrics for formal chatbot evaluation

D. Improvements Implemented

Iterative refinement focused on three areas: content quality, response presentation, and retrieval robustness. The corpus was expanded to cover missing details (contact points, financial documentation, SOLAR procedures, graduation deferral). Links in answers were standardized to official web pages rather than static PDFs to ensure recency and clarity. Fallback messages were rewritten with explicit next-step guidance, and retrieval parameters were adjusted to maintain context across multi-part queries. Indicatively, these changes improved observed accuracy from about 76 percent in early tests to roughly 88 percent in later cycles, with the largest gains in curriculum and application-process questions.

E. Remaining Limitations

Minor rendering issues may persist when responses originate from complex PDFs, though reliance on cleaned FAQ and website text reduces their frequency. Limited configurability of fuzzy matching can still affect handling of unusual typos or terse queries. Finally, the current platform lacks a built-in analytics dashboard for live monitoring of popular intents and unresolved queries, motivating consideration of an external analytics pipeline in subsequent phases.

VI. CHALLENGES

The development of the NUS MSBA chatbot faced several technical, operational, and testing challenges, each shaping its final design and implementation.

A. Technical Challenges

One of the earliest hurdles was the quality and structure of source documents. While the MSBA website and brochure contained a wealth of information, much of the brochure content originated from PDFs with formatting inconsistencies, irregular spacing, and embedded artefacts. These issues disrupted clean text extraction particularly for numeric data such as tuition fees and dates which often required additional preprocessing to ensure accurate display in the chatbot's responses.

Another recurring technical challenge was maintaining response accuracy while mitigating hallucination risks. Since the chatbot used a Retrieval-Augmented Generation (RAG) framework, the retrieval stage had to be tuned precisely to consistently return the most relevant text chunks from the knowledge base. In some cases, the system retrieved partially relevant but incomplete content, producing factually correct yet insufficient answers. Addressing this required:

- Adjustments to chunking strategies
- Updates to embeddings
- Fine-tuning prompt templates for full-context usage

B. Operational Constraints

A key operational requirement was that the system be maintainable by non-technical staff after deployment. This significantly limited the use of highly customized, code-intensive solutions that would require ongoing engineering support. While open-source frameworks could provide more flexibility, they also demanded greater technical expertise for maintenance.

This constraint reinforced the decision to adopt a managed platform like Google Vertex AI Agent Builder, which offers:

- A secure, enterprise-grade environment
- No-code configuration options for quick updates
- Integration with Google Cloud Storage for versioned, scalable knowledge base updates

C. Testing Limitations

The BAC testing document provided a strong foundation for evaluating the chatbot, but it could not fully capture the diversity of phrasing, ambiguity, and edge cases present in real-world

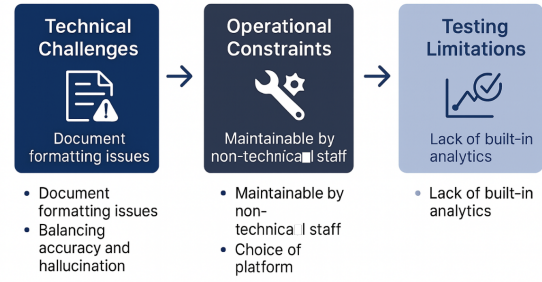


Figure 7. Challenges Overview

user queries. As a result, some knowledge gaps and response inaccuracies only emerged during iterative testing cycles.

Furthermore, the absence of a built-in analytics dashboard in Vertex AI limited post-deployment monitoring capabilities. Without detailed usage analytics, it was challenging to track query trends, identify underperforming responses, or quantify accuracy improvements over time. To address this, the team can explore implementing alternative logging mechanisms and user feedback capture processes, ensuring that performance issues could still be detected and acted upon in future updates.

VII. HANDOVER

The handover phase is designed to ensure that the NUS MSBA programme office can independently manage, update, and monitor the chatbot after the project's completion. A comprehensive handover package to be handed over includes a step-by-step maintenance guide, annotated screenshots of the Vertex AI Agent Builder interface, and a document mapping all uploaded sources to their corresponding knowledge base entries. The guide details the process for adding new documents, removing outdated content, modifying fallback responses, and updating prompt configurations.

Particular attention is given to the model context setup, which plays a central role in defining the chatbot's personality, tone, and scope of responses. The handover documentation outlines the exact structure of the model context prompts, including the balance between formal institutional tone and conversational accessibility, as well as examples of how prompt wording impacts response style.

Training sessions are to be planned with designated staff members to familiarize them with both routine and exceptional maintenance scenarios. This includes handling content updates triggered by changes in admission policies, curriculum revisions, or website redesigns. The training is to also cover the integration process for embedding the chatbot widget into the MSBA website, along with guidelines for updating domain restrictions to preserve security.

The handover process if required can incorporate a review of limitations and a roadmap for potential enhancements, such as improving fuzzy matching, integrating analytics dashboards, and expanding the knowledge base with alumni testimonials or career outcome data. This is to ensure that the receiving team

understands not only how to operate the chatbot but also how to strategically evolve it in the future.

VIII. CONCLUSION

The NUS MSBA chatbot project demonstrated the feasibility and value of deploying a Retrieval-Augmented Generation solution within a higher education admissions context. By combining a well-curated document corpus with the capabilities of Google Vertex AI Agent Builder, the system was able to deliver accurate, traceable, and conversational responses to a diverse range of prospective student queries. The integration of source citations and controlled fallback behavior further enhanced trustworthiness, a critical factor in academic communications.

Beyond its immediate utility as an information delivery tool, the project serves as a case study in responsible generative AI adoption. The careful attention paid to grounding, citation, and scope restriction ensured that the system remained aligned with institutional values and avoided the pitfalls of over-reliance on unconstrained language models. The iterative testing process, informed by structured evaluation criteria and user feedback, allowed for measurable improvements in accuracy and user experience over successive cycles.

Operationally, the choice of a managed, low-code platform facilitates a smooth handover, enabling non-technical staff to maintain and update the chatbot without requiring ongoing developer support. This ensures the sustainability of the solution while allowing for flexibility in adapting to future programme changes.

Looking ahead, the chatbot's foundation can support further enhancements, such as integrating live event schedules, expanding multi-language capabilities, and incorporating conversational analytics for deeper insight into user behavior. These developments could transform the chatbot from a primarily informational tool into a dynamic engagement platform that supports the MSBA programme's broader recruitment and retention strategies.

In conclusion, this project not only delivered a functional and well-tested chatbot solution but also established a scalable framework for integrating AI-driven conversational interfaces into the university's digital ecosystem. By addressing both the technical and operational dimensions of deployment, it provides a model for similar institutions seeking to modernize their outreach and improve prospective student engagement.

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APPENDIX

APPENDIX A: SUPPLEMENTARY SCREENSHOTS

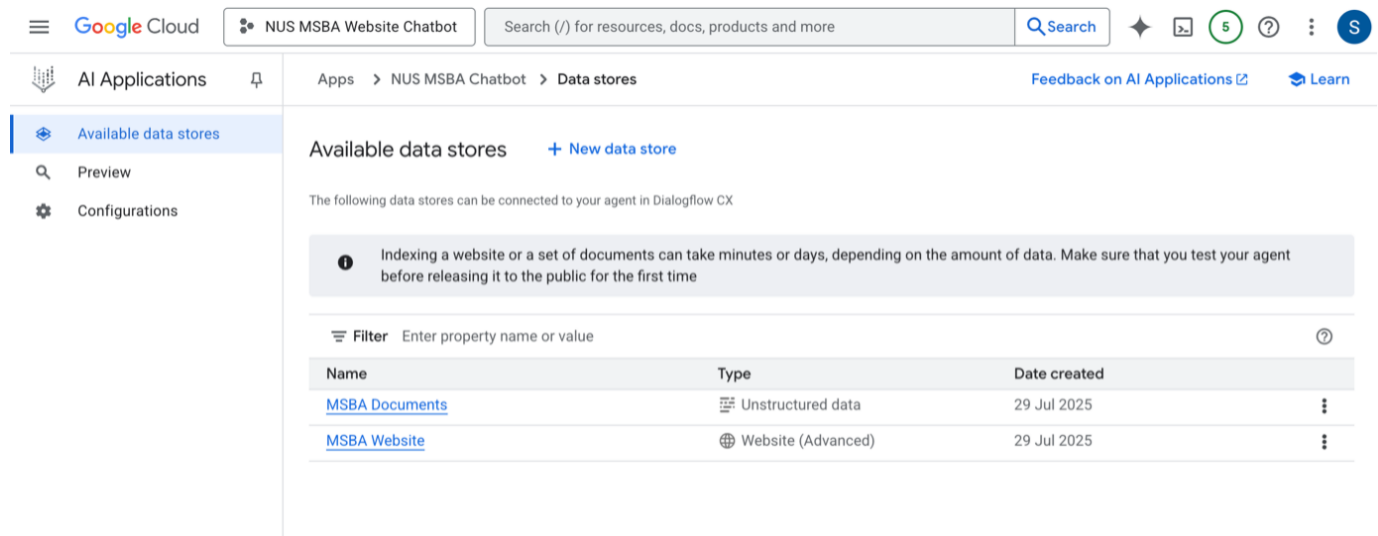


Figure A1. Google Cloud Storage (GCS) Bucket Structure

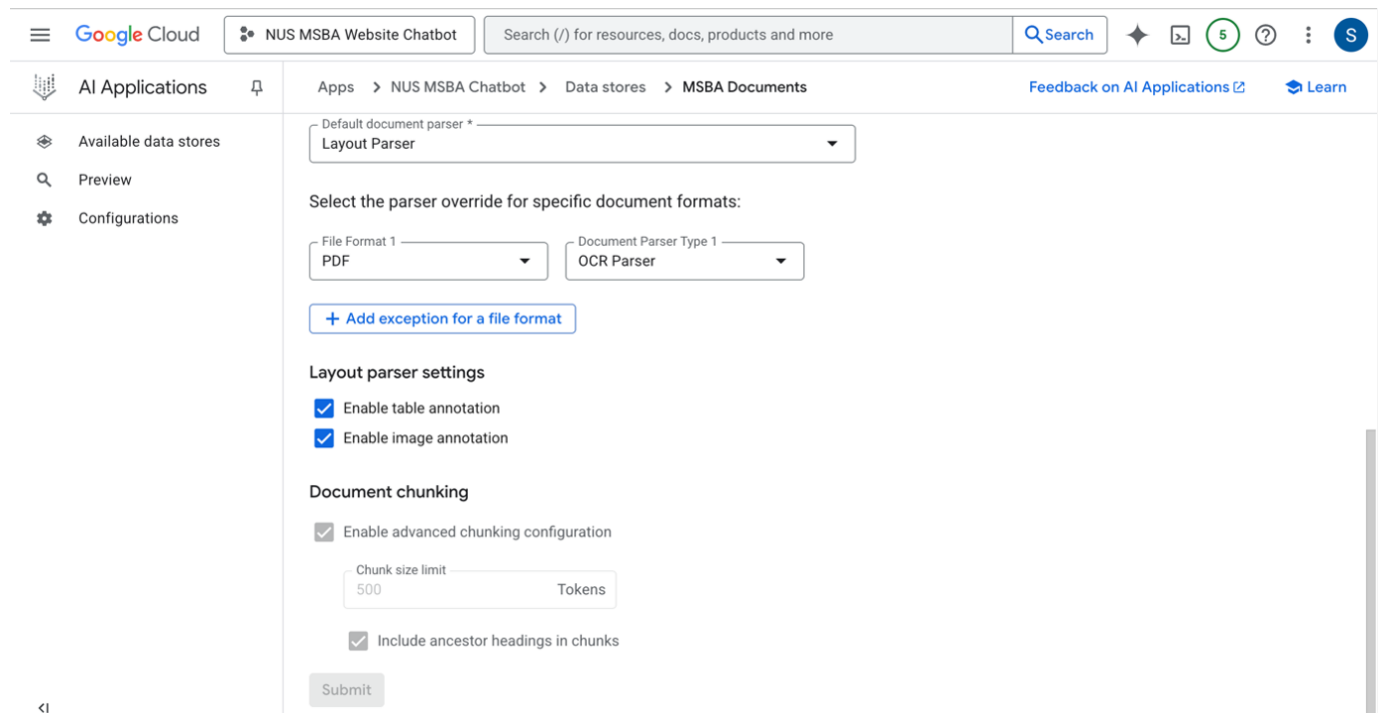


Figure A2. Chunking and Embedding Configuration

The screenshot shows the 'Settings' page for a conversational agent, specifically the 'Generative AI' tab. The interface includes a top navigation bar with 'Project' (NUS MSBA We...), 'Agent' (NUS MSBA Ch...), and 'Language' (en) dropdowns. The 'Settings' page has tabs for General, Generative AI (selected), Deterministic Flows, Connectivity, Speech and IVR, UI Settings, and Security. The 'Your prompt:' section contains the following text: 'Your name is NUS MSBA Virtual Assistant, and you are a helpful and polite You are an intelligent and helpful assistant for NUS MSBA program at NUS MSBA, You provide concise, professional, and factually accurate answers. Your task is to assist humans Always reference only the information from the retrieved content.' Below this, there are input fields for 'Agent name' (NUS MSBA Virtual Assistant), 'Agent identity' (You are an intelligent and helpful assistant for NUS MSBA program), 'Company name*' (NUS MSBA), 'Company description' (You provide concise, professional, and factually accurate answers), and 'Agent scope' (Always reference only the information from the retrieved content). The 'Data store model selection' section shows a dropdown for 'Summarization model' set to 'gemini-2.0-flash-001'.

Figure A3. Model Context Prompt Configuration

The screenshot shows the 'Settings' page for a conversational agent, specifically the 'Generative AI' tab. The interface includes a top navigation bar with 'Project' (NUS MSBA We...), 'Agent' (NUS MSBA Ch...), and 'Language' (en) dropdowns. The 'Settings' page has tabs for General, Generative AI (selected), Deterministic Flows, Connectivity, Speech and IVR, UI Settings, and Security. The 'Generative fallback' section is active, showing a 'Template' dropdown set to 'Fallback message' with 'Edit' and 'Delete' buttons. The 'Text prompt' field contains the following text: 'You are a friendly AI assistant that helps humans with \$flow-description. In particular, you can \$route-descriptions. The conversation between the human and you so far was: \${conversation USER:"Human:" AGENT:"AI"} Then, the human asked: \$last-user-utterance You say: Thank you for your question. We're sorry, but we're unable to provide an answer to this enquiry at the moment. You may wish to email your query to msba@nus.edu.sg and the team will be happy to assist you further.'

Figure A4. GenAI Fallback Message Setting

Project: NUS MSBA We... Agent: NUS MSBA Ch... Language: en

Integrations

Text Based

Active

Conversational Messenger

Manage

Slack

Connect

Conversational Messenger

Conversational Messenger brings a rich UI for Conversational Agents that enables developers to easily add conversational agents to websites. [View Documentation](#)

You can also, modify the interactive visual interface, by going to Agent settings then click on Multimodal, and then click on [UI settings](#)

! End-user interactions with the Conversational Messenger widget will be billed to your GCP account.

Add this agent to your website by copying the code below:

```
<link rel="stylesheet" href="https://www.gstatic.com/dialogflow-console/fast/df-messenger/prod/v1/themes/df-messenger-default.css">
<script src="https://www.gstatic.com/dialogflow-console/fast/df-messenger/prod/v1/df-messenger.js"></script>
<df-messenger
  project-id="nus-msba-website-chatbot"
  agent-id="9aa0d87f-1c37-428f-82cb-aa356beabcc1"
  language-code="en"
>
```

Environment: Draft

Disable Conversational Messenger New session Done

NUS MSBA Chatbot

i do not have my degree cert now, how do i proceed with admission

To apply for the MSBA program without your degree certificate, apply via Graduate Admission System (GDA3). Ensure you have all documents prepared by checking the Application Checklist. You can find the application at <https://msba.nus.edu.sg/apply-now/>.

Website Chatbot FAQs

Website Chatbot FAQs

Admission and Application Will applications without GRE or GMAT scores still be considered...

Ask something...

Figure A5. Citation Transparency

Project: NUS MSBA We... Agent: NUS MSBA Ch... Language: en

Settings

Save Cancel

General

Generative AI Deterministic Flows Connectivity Speech and IVR UI Settings Security

Enable conversation history

☒ Indicates whether you would like Google to collect and store redacted end-user queries for quality improvement. Note: some features, such as [Experiments](#), export to BigQuery and NLU model improvement, require this setting to be enabled, as they rely on traffic logs. Redaction needs to be configured by the developer through [DLP](#) or [parameter redaction](#).

Specify a Cloud Storage URI to store and playback voice conversations

Google Cloud Storage URI

gs://bucket/object-prefix-or-name

[Browse Google cloud storage](#)

☐ Enable consent-based end-user input redaction

Allow consent-based redaction behavior on end-user input and parameters based on special session parameter. [Learn more](#)

BigQuery export

☐ Enable BigQuery export

Export conversation history to BigQuery. [Learn more](#)

Intent suggestions

Receive suggestions to improve existing intents, and create new intents.

☐ Enable intent suggestions

User feedback

Provide feedback about the quality of the agent's answers which includes thumbs up/down.

☒ Enable user feedback

Figure A6. Conversational Logging and User Feedback Setting

APPENDIX B: EXTERNAL FIGURES REFERENCED

- 1) Figure 2. RAG Workflow Process: <https://www.mygreatlearning.com/blog/enhance-large-language-models-with-rag/>
- 2) Figure 3. GenAI workflow on Vertex AI: <https://cloud.google.com/vertex-ai?hl=en>